

Homework 3: Nico Boyle, Owen Moorhus, Asher Olgeirson

1. Consider the following proposition, which we could label as $P(n)$.
If $5n + 3$ is even, then n is odd.

Prove by strong induction that for every positive integer n , proposition $P(n)$ is true.

Proof :

By strong induction, assume that $P(k)$ is true for $1 \leq k \leq n - 1$

The base case of $k = 1$ holds as $P(1) = 8$ which is even, and 1 is odd.

As does the base case of $P(2) = 13$, which is odd, and 2 is even.

$$5(n - 2) + 3 = (5n + 3) - 10$$

As $5n + 3$ is even, as is 10, $5(n - 2) + 3$ is therefore even.

This proves the inductive hypothesis as $P(n - 3) \implies P(n - 2)$ ■

2. Prove by strong induction that for every positive integer n the following holds:

$$\sum_{k=1}^n k(k+1) = \frac{n(n+1)(n+2)}{3}$$

proof:

By strong induction, assume:

$$\sum_{k=1}^{n-1} k(k+1) = \frac{n(n+1)(n-1)}{3}$$

The base case $n = 1$ holds:

$$\sum_{k=1}^1 k(k+1) = 1(1+1) = 2 = \frac{1(1+1)(1+2)}{3}$$

By substituting the last term of the summation:

$$\sum_{k=1}^n k(k+1) = \frac{n(n-1)(n+1)}{3} + n(n+1)$$

The inductive hypothesis is proved below:

$$\begin{aligned} \frac{n(n-1)(n+1)}{3} + n(n+1) &= \frac{n(n+1)(n+2)}{3} \\ n(n-1)(n+1) + 3n(n+1) &= n(n+1)(n+2) \\ n^3 - n + 3n^2 + 3n &= n^3 + 3n + 2n \\ n^3 + 3n^2 + 2n &= n^3 + 3n^2 + 2n \end{aligned}$$

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